

Game Name

Survivor Z (3D)

New Feature 1: Melee Combat System (Sword)

Description

In addition to guns and bombs, the player can use a melee sword to attack zombies at close range. This allows instant damage without ammo consumption and supports aggressive, skill-based combat.

Melee Mechanics

- Short-range attack arc in front of the player
 - Damage applied using collision or hitbox detection
 - Faster attack speed than guns
 - No ammo required
- High risk due to close proximity to enemies

Gameplay Impact

- Encourages close-combat playstyle
 - Useful when ammo is low
 - Complements Player 1's aggressive role
 - Adds combat variety and intensity
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New Feature 2: Temporary Weapon Pickup System

Description

Special guns spawn randomly in the arena and can be picked up by the player for a limited duration.

Key Rules

- Temporary weapons override the current gun
- Each weapon has a usage duration timer
After expiration, the previous gun is restored
- Weapons cannot be stored
- HUD shows remaining usage time

Gameplay Impact

- Encourages movement and exploration
- Creates short-term power spikes
- Adds time-based tactical decisions

New Feature 3: Dual Player Switching System

Description

The game includes two switchable characters with different strengths. The player can switch between them at any time.

Player Types

Attribute	Player 1 – Human (Offensive)	Player 2 – Superhuman (Defensive)
Health	Low	High
Weapon Damage	High	Low
Playstyle	Aggressive	Tanky
Movement	Faster	Slightly slower
Ideal Use	Wave clearing	Survival & defense

Switching Mechanics

- Instant switching with cooldown
 - Only one active player
 - Inactive player:
 - Becomes invulnerable
 - Gradually regenerates health
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Game Overview & Features

Survivor Z is a 3D OpenGL-based survival game where the player is placed in a bounded arena and must survive continuous waves of incoming zombies. Zombies spawn dynamically along arena edges and pursue the active player using vector-based following logic. The player can aim and shoot guns to eliminate enemies, while collision detection ensures accurate hits and reduces player life upon enemy contact.

The game includes cheat mode, progressive enemy speed, headshot detection, score multipliers, Corpse Tactics System, gun-based combat, a plantable Time Bomb, melee combat using a sword, a temporary weapon pickup system, dual-player switching, and a teleportation ability for strategic repositioning.

The Time Bomb detonates after a fixed delay, killing enemies within a defined radius—including the player if caught in the blast. Melee combat allows close-range sword attacks, encouraging aggressive positioning and risk–reward gameplay.

The arena, enemies, bombs, bullets, and melee interactions are fully procedural, demonstrating 3D transformations, camera projection, and time-based game logic. The game ends when the player’s life reaches zero.

Weapon System Details

1. Guns

Type: Ranged

Mechanics:

- Shoots bullets using ray or projectile logic
- Ammo-based shooting
- Affected by cheat mode (auto-aim)

Purpose:

- Primary combat method
 - Precision-based gameplay
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2. Time Bomb

Type: Tactical / Area-of-Effect

Description:

The player can plant a Time Bomb at their current position. After a fixed countdown, the bomb explodes, dealing massive area damage.

Rules & Behavior:

- Activates after a countdown timer
- Circular explosion radius
- Instantly kills zombies inside the radius
- Player takes damage or dies if inside the blast
- Limited usage or cooldown

Key Concepts Implemented:

- Time-based events
 - AoE collision detection
 - Friendly fire logic
 - Risk-reward decision-making
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Updated Feature Table

Feature	Description	Key Mechanics
Cheat Mode	Auto-aim & firing	Auto-rotation
Increasing Enemy Speed	Difficulty scaling	Time-based logic
Headshot Detection	Instant kills	Separate hitboxes
Score Multiplier	Rewards accuracy	Combo logic
Corpse Tactics System	Environmental traps	Chain reactions
Temporary Weapon Pickup	Time-limited guns	Timers
Dual Player System	Two characters	Health regen
Teleportation	Emergency escape	Safe reposition
Melee Sword System	Close-range combat	Hitbox detection

Updated Work Distribution

Member 1 – Graphics & Environment

- Design 3D arena, walls, and grid floor
- Create zombie models with head/body hitboxes
- Model guns, Time Bomb, and sword weapon
- Design Player 1 & Player 2 models
- Implement explosion and melee visuals
- Handle camera and OpenGL rendering

Member 2 – Game Logic & Mechanics

- Implement zombie AI and speed scaling
- Implement gun shooting, bomb explosion, and melee collision logic
- Program cheat mode, teleportation, and Time Bomb behavior
- Implement Corpse Tactics System

- **Handle temporary weapon timers**
- **Implement player switching and stat differences**

Member 3 – Gameplay & Scoring System

- **Implement health systems for both players**
- **Implement health regeneration for inactive player**
- **Implement score multipliers**
- **Handle weapon switching (guns ↔ sword ↔ bomb)**
- **Manage interactions with power-ups and bombs**
- **Implement game states**